



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Artificial Intelligence and Data Science

Academic Year: 2024- 2025 (ODD Semester)

Active Learning Best practices: Flipped Classroom – Virtualization

Degree, Semester & Branch: V Sem. B.Tech. Artificial Intelligence and Data Science

Course Code & Title: - CCS335 – Cloud Computing

Name of the Faculty member: Mr. P. VETRIVEL, AP/AI & DS

Theme of discussion: Flipped Classroom

Topics Covered: Unit- II

Date and Time: 09.09.2024 & 01.55 P.M to 02.40 P.M

Active Learning Best practices: Flipped Classroom – Virtualization

Unit / Topic: Unit – II / Virtualization

Course Outcome: CO2 - To understand the concepts of virtualization and virtual machines.

Topic Learning Outcome: TLO08

Activity Chosen: Flipped Classroom

Justification:

- Virtualization in cloud computing used to optimizing resource usage, reducing hardware requirements, and enhancing scalability. It allows multiple virtual machines (VMs) to run on a single physical server, maximizing resource efficiency and lowering both capital and operational expenses. Virtualization also enables on-demand scaling, better multitenancy, and dynamic resource allocation, reducing the need for excess physical infrastructure. Additionally, it improves disaster recovery, enhances high availability, and lowers energy consumption, all contributing to cost savings for cloud providers and customers.

Time Allotted for the Activity: 30 Minutes

Details of the Implementation:

The flipped classroom activity was implemented as a teaching strategy to enhance student engagement, foster deeper understanding, and promote active learning. In a traditional classroom, students typically receive lectures during class time and do homework or assignments outside of the class. However, in a flipped classroom, this model is reversed - students watch pre-recorded lectures or read material at home and On the day of activity ask the students to explain about the concept. This report summarizes the implementation and outcomes of the flipped classroom activity.

Virtualization:

Virtualization is the process of creating a virtual version of resources such as servers, storage, or networks, allowing multiple virtual instances to run on a single physical system. It enables efficient use of hardware, isolation, scalability, and cost savings in cloud computing and data center management.

Types of Virtualizations:

1. Hardware Virtualization (Server Virtualization):

- Definition: Divides a physical server into multiple virtual machines (VMs), each with its own OS and resources. This allows better hardware utilization and flexibility.
- Hypervisor Types:
 - Type 1 (Bare-metal): Runs directly on hardware (e.g., VMware ESXi, Microsoft Hyper-V).
 - Type 2 (Hosted): Runs on top of an existing OS (e.g., Oracle VirtualBox, VMware Workstation).

2. Storage Virtualization:

- Definition: Combines multiple physical storage devices into a single virtual storage pool. It abstracts storage management, making it easier to manage and scale storage resources.
- Example: Storage Area Networks (SAN) and Network Attached Storage (NAS).

3. Network Virtualization:

- Definition: Abstracts physical networking hardware into multiple virtual networks, allowing multiple virtual network devices (routers, switches) to operate on a single physical network infrastructure.
- Example: VLANs (Virtual Local Area Networks), Software-Defined Networking (SDN).

4. Desktop Virtualization:

- Definition: Allows users to access desktop environments remotely, with the desktop hosted on a centralized server rather than individual devices.
- Example: Virtual Desktop Infrastructure (VDI), Remote Desktop Services (RDS).

5. Application Virtualization:

- Definition: Separates applications from the underlying operating system, allowing them to run on any device without being installed on the local machine.
- Example: Citrix Virtual Apps, Microsoft App-V.

Correctness

The correctness of virtualization refers to the ability of a virtualized environment to accurately and reliably simulate physical resources, ensuring that virtual machines (VMs) and virtualized services function as intended without errors, inefficiencies, or security risks. Correctness in virtualization involves several key aspects:

- Resource Isolation and Security
- Accurate Resource Allocation
- Compatibility
- Performance Consistency

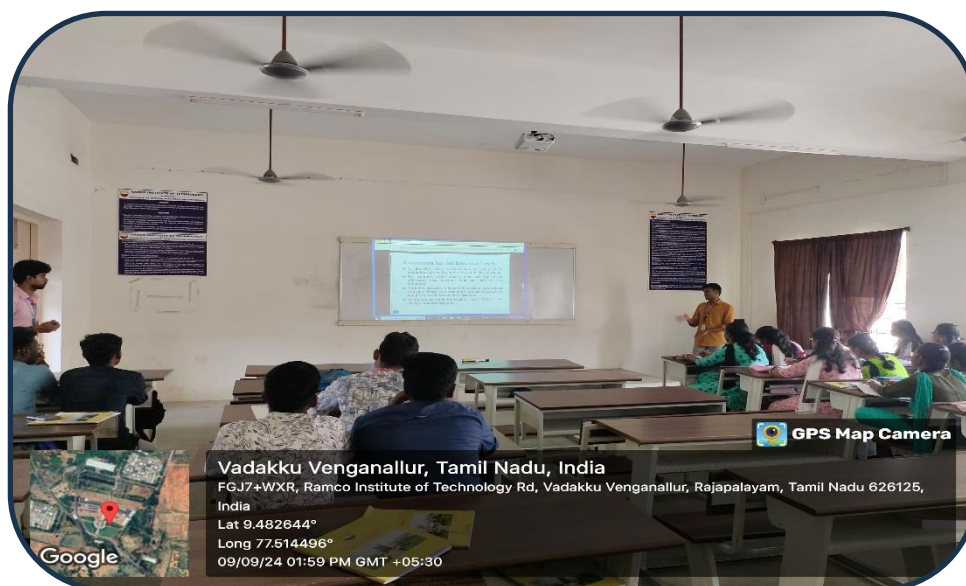
CO – PO / PSO Mapping

CO	PO1	PO2	PO5	PO9	PO10	PO11	PO12	PSO3
CO2	3	2	2	2	2	2	2	3

PO/PSO Mapped

- **PO1:** Students can be able to apply the principles of computer science and engineering to solve a virtualization issues.
- **PO2:** Students can be able to compare various virtualization types to identify the limitations.
- **PO5:** Students can be able to identify modern engineering tools, techniques and resources to create a virtual machine.
- **PO9:** Students can be able to demonstrate effective communication, problem-solving skills by apply the concept of virtualization and its types to create the virtual machine.
- **PO10:** Students can be able to create engineering-standard figures, reports and drawings to create a virtual machine.
- **PO11:** Students can be able to analyze the concept of virtualization and it types to evaluate an engineering project.
- **PO12:** Students can be able to recognize the need and be able to apply the virtualization concept and its type for new development of cloud computing.
- **PSO3:** Students will be able to enrich the critical thinking skills in virtualization concept to foresee the research findings and provide the solutions.

Glimpses:



Reflective Critique:

- **Preparing and Sharing Content:** Video lectures and reading materials were created covering the relevant topics. These materials were shared with the students through the learning management system or other suitable platforms well in advance of the corresponding class sessions.
- **Class Sessions:** During the class sessions, students were asked to discuss and apply the concepts covered in the pre-recorded lectures. Facilitators were present to guide and support the students' learning process.
- **Active Learning Activities:** Students are asked to explain about the concept in front of the classroom, and peer teaching sessions were conducted.
- **Feedback Collection:** Students were asked to provide feedback on the flipped classroom experience, highlighting the aspects they found beneficial and suggesting potential improvements.

Observations:

The implementation of the flipped classroom activity was successful in enhancing student engagement, promoting deeper understanding, and creating a more interactive learning environment. By using this approach, students were encouraged to take ownership of their learning process and actively participate in class activities. However, continuous refinement based on student feedback and ongoing assessment of the approach will be essential to optimize its effectiveness in future implementations.

Benefit of the practice:

- The students understand well about virtualization Concept.
- The students can able to Interact with the faculty member and their class mates during the class was effective.
- It helps the students to improve the self-learning of a student.
- With the help of this activity, each student can get clear idea about the use of snapshot algorithms in distributed systems.

Students Response:

- Bright students were actively participated to prepare the topic wise questions and concept notes.
- Slow learners expect to assist the topic towards searching online and also understanding the questions.
- Most of the students were enjoyed the session.
- Students response ensures the students could improve listening, communication, creativity and problem-solving skills.


Signature of the Faculty Member


HoD